Assignment: Needs-based Project Scenario [Group Component]

[Max. Marks = 300, Weight = 15%]

Part-A: Your Project's Proposed Solution Block Diagram IoT Architecture Design

1. Describe the use case for your project; include needs (problem/pain of the customer / sponsor), criteria, and constraints / limitations as set / expected by the customer / sponsor. [50 Marks]

Stakeholder(s), i.e., customer / sponsor {10 Marks}

The main stakeholders are power grid companies and grid operators, as they are responsible for distributing safe and reliable electricity. Secondary stakeholders include municipal governments, emergency services, homeowners, and general grid users.

Pain/Problem/Opportunity {10 Marks}

Many grids today lack a huge thing: data. They lack sensors all over equipment to detect signs of failure before they happen; they lack sensors that detect the locations of faults so blackouts last hours as utilities has to find them, they do not have sensors in remote locations to monitor signs of high demand, which can cause problems for the grid. These issues cause longer outages and higher maintenance costs.

IoT-based Solution {20 Marks}

Aegis Grid is designed to improve power grid monitoring and reliability by collecting and analyzing real-time data. This system will have sensors placed on transformers, power lines, generators, and so on to continuously measure voltage, current, temperature, humidity, and vibration.

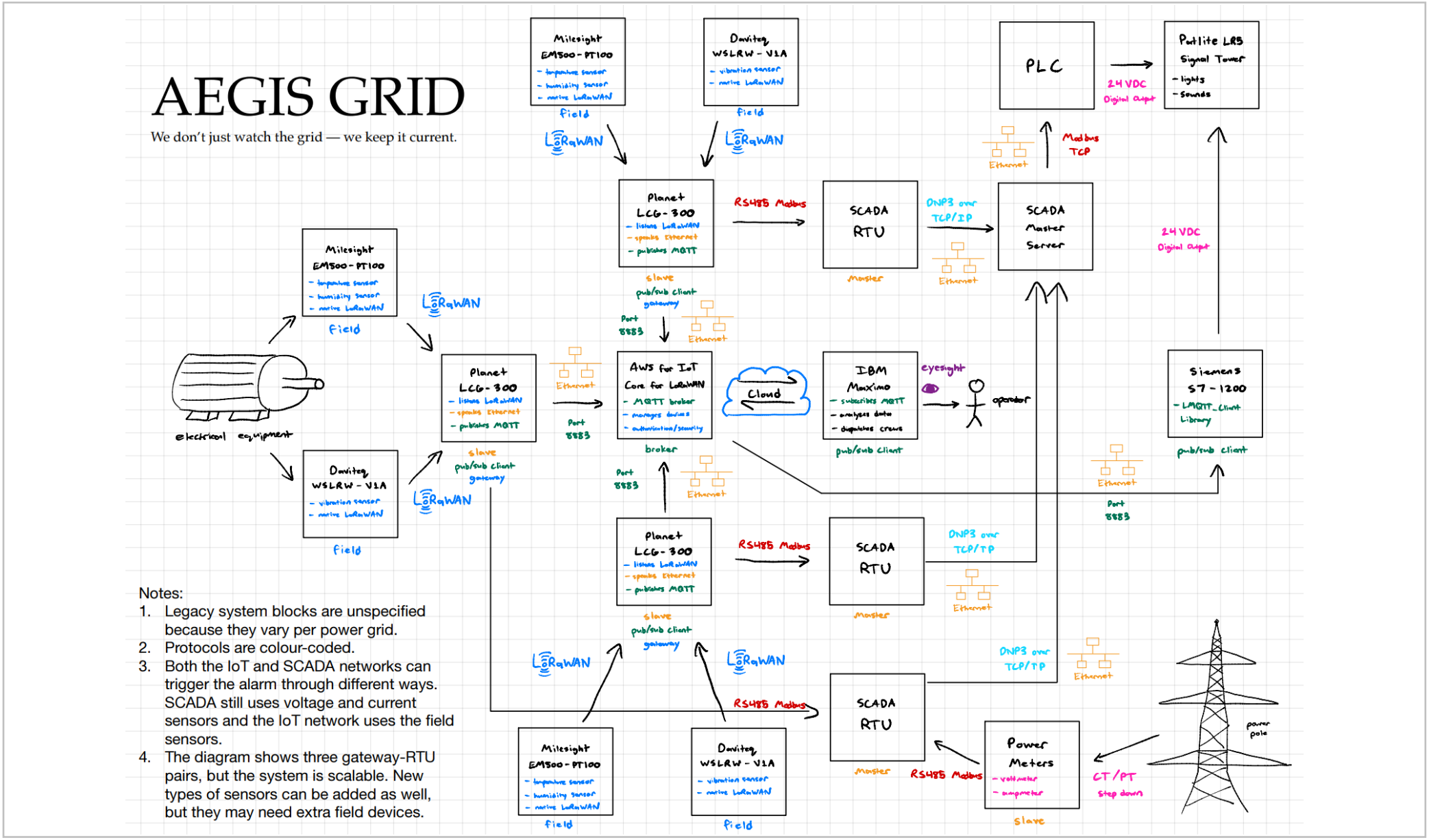
These sensors will use LoRaWAN. The gateways will then send data to the cloud to be analyzed and monitored by operators. The reason we chose LoRaWAN for this application is because power grids span entire landscapes and we need a protocol that is built for long range, and the data we send may come from sensors from different manufacturers that may only be compatible with their own field device or microcontroller, and so the point to multipoint standard of LoRaWAN allows for multiple field devices so that these sensors only need to be compatible with LoRaWAN to work.

Sensors will be placed on equipment to detect signs of impending failure. Sensors will be placed on each feeder line to detect signs of high demand spikes. Sensors will be placed along each feeder line to detect locations of faults if they do occur.

Value Creation {10 Marks}

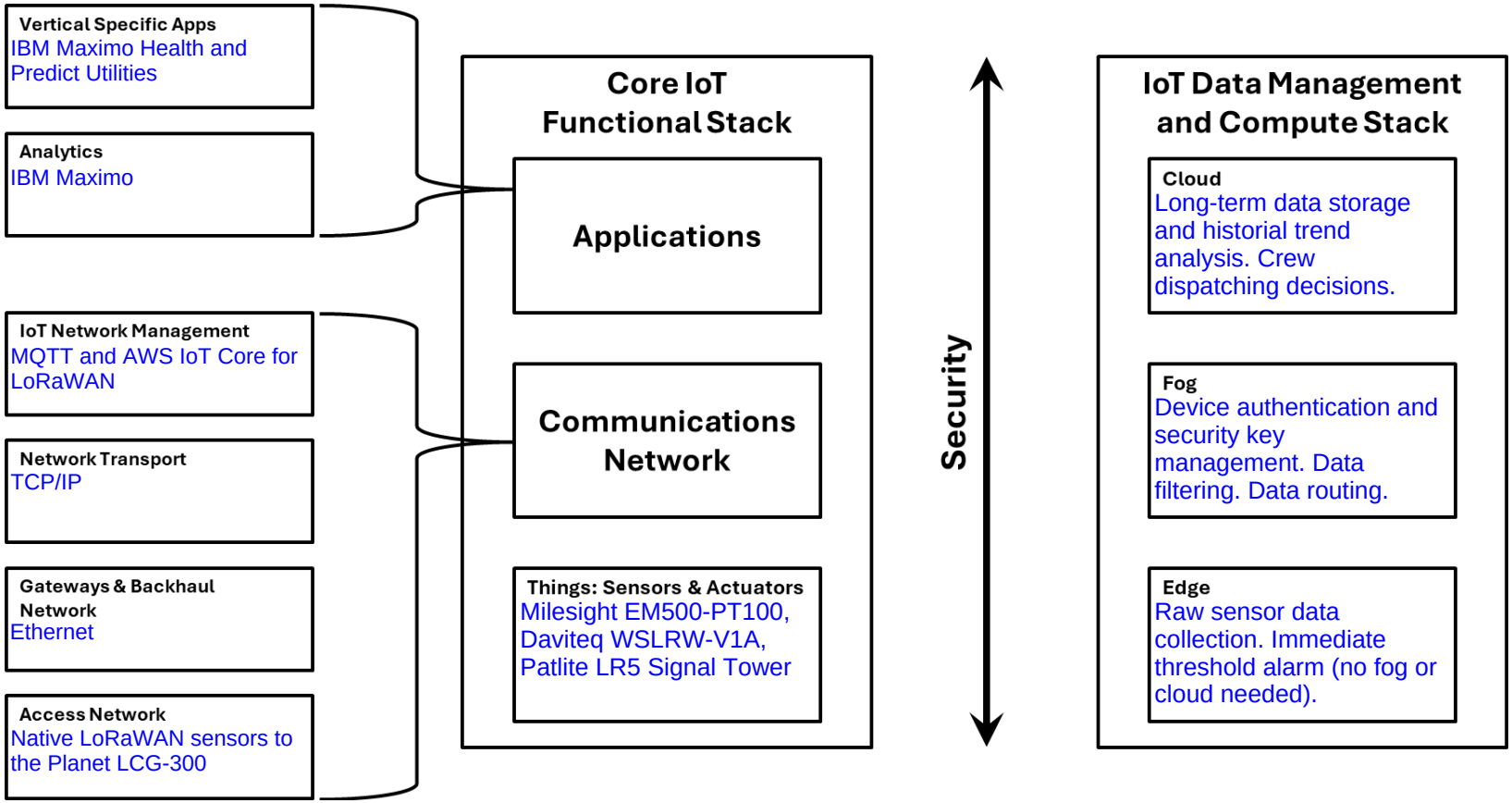
The core problem being solved is information latency and defiency in the power grid. Right now, utilities are running blind in many areas--they find out something is wrong only after it has already failed. The value of this system shifts the reactive to predictive.

2. a. Draw a block diagram for your project's proposed solution. [40 Marks]



Notes:
1. Legacy system blocks are unspecified because they vary per power grid.
2. Protocols are colour-coded.
3. Both the IoT and SCADA networks can trigger the alarm through different ways. SCADA still uses voltage and current sensors and the IoT network uses the field sensors.
4. The diagram shows three gateway-RTU pairs, but the system is scalable. New types of sensors can be added as well, but they may need extra field devices.

b. Design the Simplified IoT Architecture: Expanded View for your project's proposed solution; include both the Core IoT Functional Stack and the IoT Data Management and Compute Stack. [50 Marks]



Feel free to elaborate in the space below, if needed.

For Aegis Grid's feature of demand spike monitoring, we are assuming this is already done by the legacy systems in place (SCADA, voltmeters and ampmeters after current transformers (CTs) and potential transformers (PTs) and RTUs). It will still be demonstrated in the proof-of-concept and the additional feature to this system Aegis Grid will offer the LoRaWAN and MQTT cloud aspect. IBM Maximo will handle analytics, visualization, and processing at the cloud layer and will make decisions to dispatch crews based on data, but there will be hard threshold alarms for emergency conditions at the edge layer that will immediately sound alarms at the nearest substation or central site. The sensors we chose have a built-in LoRaWAN unit in them so they don't actually need to be connected to a field device microcontroller. They access the network through LoRaWAN, send their data to the gateway on site, the gateway converts the data from LoRaWAN to ethernet connected with a wire to the RTUs or PLCs of the legacy systems, and publishes to an MQTT server as well. Some SCADA systems are modernized and can subscribe to topics, but if not, IBM Maximo will. AWS IoT Core for LoRaWAN handles device authentication, security, sensor and gateway additions, and general management of the devices.

c. For the drawing and the design, include and justify the selection of devices, designing of networks, and choosing of protocols based on needs, criteria, and constraints / limitations as set / expected by the customer / sponsor. [10 Marks]

Milesight EM500-PT100: industrial-grade, 10 year battery life, for harsh outdoor environments, native LoRaWAN support
Daviteq WSLRW-V1A: industrial, frequency response over 10kHz for specific fault signature like bearing wear and mechanical imbalance, native LoRaWAN support
LoRaWAN: power grids span large geogrpahic areas and it is long range, wide range of sensors may be incompatible with the same field/gateway devices and it is point to multipoint
Planet LCG-300: One of the only gateway devices that speaks LoRaWAN on the sensor side and both MQTT over Ethernet and Modbus TCP on the network side for legacy systems such as SCADA and PLCs
SCADA: not IN the IoT network, but retained because power grids already run on them
Patlite LR5 Signal Tower: standard lights and sounds for an alarm signal (can pick any)
AWS IoT Core for LoRaWAN: Fully managed service for a LoRaWAN network
IBM Maximo Health and Predict Utilities: dominant platform for IoT analytics for utilities

3. a. What might be the full-scale implementation/deployment budget for your project? Support with evidence. [40 Marks]

Milesight EM500-PT100: 5 x ~$170 = ~$850 (solution.seeedstudio.com, estimates)
Daviteq WSLRW-V1A: 5 x ~$150 = ~$750 (estimates, no public price)
Planet LCG-300: ~$500 – $800 (rad.com, estimates, no public price)
Patlite LR5: ~$200 (thethingsnetwork.org, estimates, no public price)
Per-site hardware total: ~$2,300 - $2,600
Ex. deployment across 10 substations in the grid: $23,000 - $26,000
AWS IoT Core for LoRaWAN: LoRaWAN messaging is priced at $2.30 per million messages in US East. For 10 sensors transmitting every 15 minutes, that's roughly 480 messages/sensor/day x 10 sensors × 365 days = ~1.75 million messages/year - approximately $4/year per substation, or $40/year for 10 sites (aws.amazon.com)
IBM Maximo Health and Predict Utilities: $50,000 - $150,000+ per year licensing
This does not include server/network infrastructure, building overhead, or employee compensation. It seems hardware is the cheapest part of the system (as expected in IoT). Software, no. Total Year 1: ~$1.1M - $5.4M. Ongoing annual: ~$100,000-$350,000

b. Is there precedence for such spending within the stakeholder segments for whom you are proposing your solution? Support with evidence. [10 Marks]

The U.S. Department of Energy estimates power outages are costing American businesses around $150 billion per year. (https://www.energy.gov/ne/articles/department-energy-report-explores-us-advanced-small-modular-reactors-boost-grid)
In 2019, power interruptions for 82 Canadian utilities cost the economy CA$34.5 billion. (https://www.sandc.com/globalassets/sac-electric/documents/public---documents/sales-manual-library---external-view/technical-paper-100-t133.pdf?dt=638366865483706430)
Example power grid company: Hydro One: In 2024, $36.7 billion in assets, $8.5 billion annual revenues. (ca.marketscreener.com). Our IoT solution is baby-money to them.

Part-B: Your Project's Proof-of-Concept Block Diagram IoT Architecture Design

1. Describe your project's proof-of-concept. How it is addressing your project's use case in Part A #1 above. Explain how your scaled-down, lean-budget version of your proposed solution would convey to the customer / sponsor that their needs have been accommodated. Also, what deficiencies in your proof-of-concept would need to be overcome for the customer / sponsor to make a decision on the full-scale implementation / deployment of your proposed solution.

[10 Marks]

We are going to test Aegis Grid's three proposed features at a smaller scale with two custom PCBs with ESP32 microcontrollers called the MacIoT board, two LoRa EBYTE modules, a breadboard and wires, a few potentiometers, a few sensors, two LEDs, and a passive buzzer. For the software side, we will program the field and gateway devices with Arduino IDE and language, use a public MQTT server (or host a broker on our PC if they don't work), and run Node-RED for visualization.

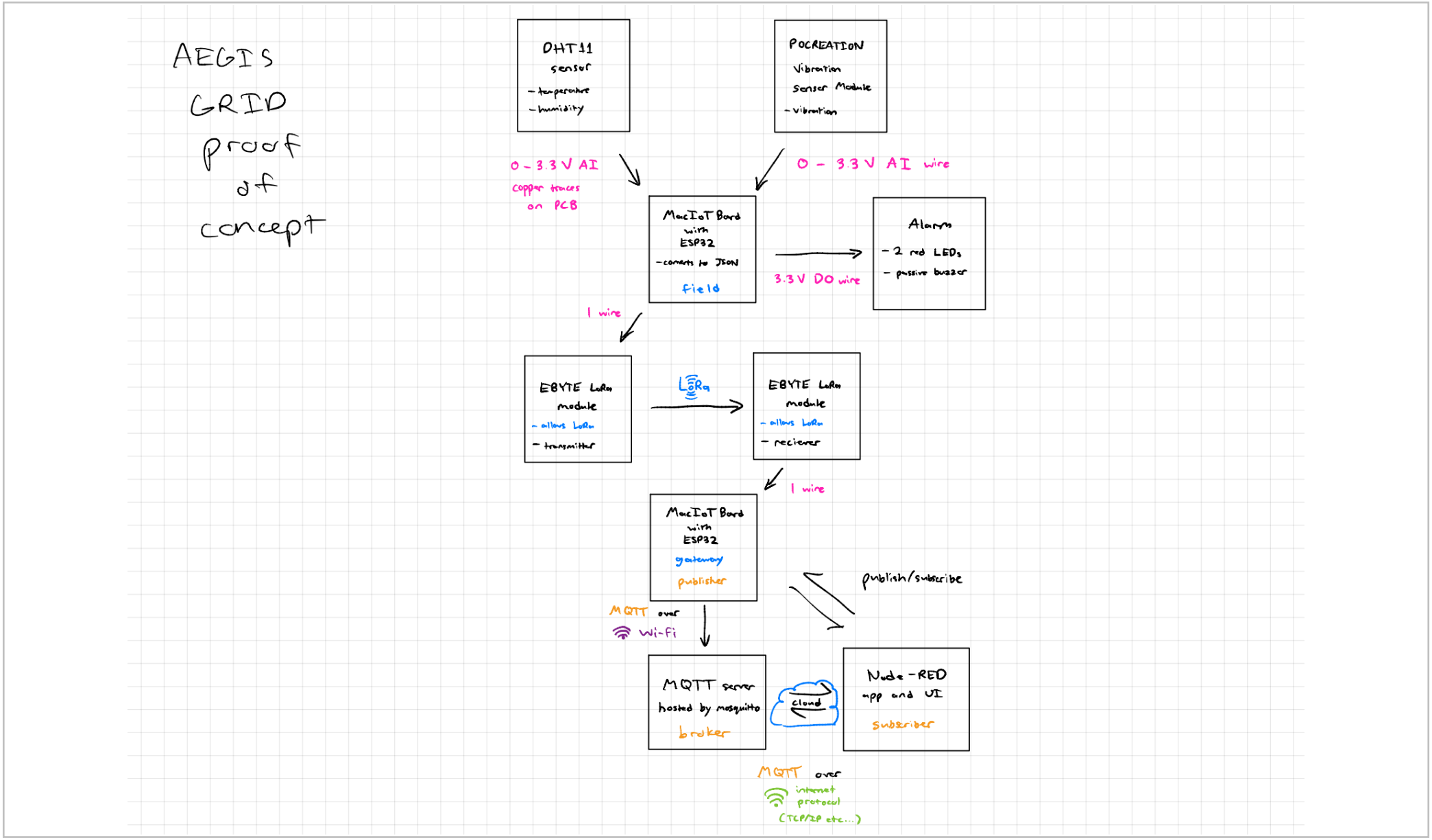
1. Demand Spike Monitoring - currently done with SCADA RTUs
Two lines of 4 resistors in series will mimic two feeder lines in the power grid. The resistors represent any load using power on that line: houses, street lights, etc. The lines run on 3.3V and have an expected voltage drop across the resistors of each line the microcontroller will measure with its analog input pins, but potentiometers can increase their voltage to 5V. If that happens, it simulates a "demand spike" in the grid, the microcontroller will sense the higher voltage, then sound the alarm and alert the dashboard. Difference with proposed solution: In real life, demand spike is measured by current draw and not voltage, even though they're related. Also, in real life, alarm decisions would be made at the cloud after data is processed, not at the edge at the field device.

2. Fault Location Detection - no current industrial equivalent, or limited equivalency
One line with 4 series resistors (so 4 nodes excluding GND) with each node wired to 4 analog input pins, measuring the voltage at each node. The microcontroller will be programmed to measure the voltage drops at each node and sound the alarm and say on the dashboard which node the fault is located at. The wires to the analog input pins represent voltage/current sensors across shunt resistors along grid feeder lines.

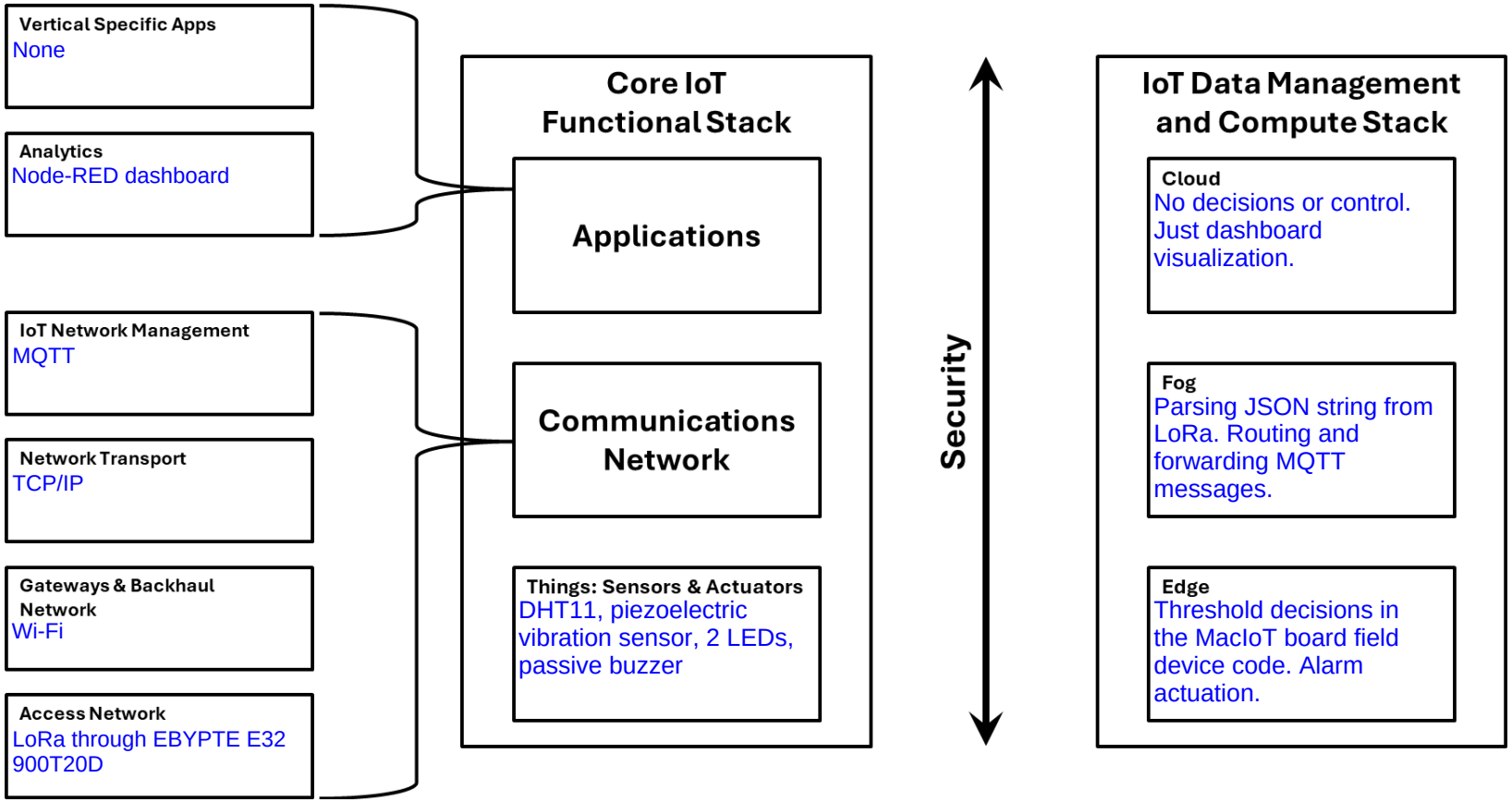
3. Equipment Preventative Maintenance - limited industrial equivalent (depends on grid)
A temperature, humidity, and vibration sensor will be connected to pins on the microcontroller to monitor the health of a small DC motor and send the data to the cloud. These sensors represent their industrial versions in the proposed solution, and the DC motor represents a piece of equipment in the power grid, ex. a generator. Thresholds for these sensors in the code, when passed, will sound the alarm. Difference with proposed solution: In real life, decision-making won't be limited to thresholds, but will be analyzed, processed, and will use statistical process control to decide if alarms should be sounded.

Access technology: LoRa. This works for our proof-of-concept because all our sensors interface directly with our control board, but our proposed solution uses LoRaWAN for the point to multipoint feature and because there may be different sensors with different compatible field devices for them.

2. a. Draw a block diagram for your project's proof of concept. [40 Marks]



b. Design the Simplified IoT Architecture: Expanded View for your project's proof-of-concept; include both the Core IoT Functional Stack and the IoT Data Management and Compute Stack. [50 Marks]



Feel free to elaborate in the space below, if needed.

Differences with proposed solution: All alarm decision-making is done at the edge, on the Arduino-written code of the field device. The cloud is essentially just visualization. The cloud also doesn't have any decision or control features, and no analytics and processing. We are using LoRa for simplicity, and because all our sensors easily interface with our MacIoT board, but our industrial solution uses LoRaWAN. Our MQTT is published over Wi-Fi, not ethernet. We don't have a vertical specific app or IoT network management software or similar. We have a field device (a custom PCB with an ESP32 microcontroller) and a separate EBYTE LoRa module that the sensors interface into the GPIO, while in our industrial solution, the sensors we chose already transmit with LoRaWAN, meaning we don't need an extra field device with GPIO that the sensors input into.